

Sara Ghasvarianjahromi, PhD

ML Researcher | Reliable Machine Learning, Information Retrieval & Distributed Learning

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Summary

Ph.D. in Electrical Engineering (NJIT, 2026) researching reliable machine learning: semantic retrieval under noisy and lossy conditions, adversarially robust decentralized learning (IEEE ISIT), and probabilistic guarantees for ML systems under uncertainty. Industry research experience at Futurewei applying deep learning and optimization to real network telemetry, published in *IEEE Networking Letters*. Hands-on with PyTorch, Hugging Face, embeddings, and statistical modeling.

Skills

Machine Learning: Semantic Retrieval & Embeddings, Deep Learning, Decentralized & Federated Learning, Adversarial Robustness, LLM Fine-Tuning (LoRA/PEFT), AI Agents, Retrieval-Augmented Generation (RAG)

Applied Domains: Network Analytics & Telemetry, QoS Optimization, Resource Allocation, Energy-Efficient Systems

Programming: Python, MATLAB, SQL (SQLite)

ML Frameworks & Libraries: PyTorch, Hugging Face Transformers, scikit-learn, pandas, NumPy

Optimization & Modeling: Gurobi, Convex Optimization, Probabilistic Modeling, Statistical Modeling

Tools: Git, Linux, VS Code, LaTeX

Experience

New Jersey Institute of Technology, NJ – Research Assistant, ML & Distributed Systems Jan 2022 – May 2026

- Built a context-aware semantic retrieval system using pretrained embeddings that preserves retrieval accuracy over noisy, lossy communication channels (paper under review, 2026).
- Designed and validated decentralized learning architectures with Byzantine-adversarial defenses in PyTorch; under attack conditions that degrade standard decentralized baselines, achieved 30–50% higher accuracy than state-of-the-art methods on MNIST and CIFAR-10, and up to 25% gains under adversarial noise.
- Implemented a fully decentralized sparse matrix multiplication framework with formal robustness guarantees, enabling scalable, coordination-free distributed computation.
- Derived probabilistic performance bounds characterizing system reliability under uncertainty.

Futurewei Technologies, NJ – ML Research Intern May 2025 – Aug 2025

- Built an end-to-end AI/ML pipeline for QoS optimization over 3GPP network analytics, reducing energy consumption by 18–25% vs. fixed-latency baselines and 40–50% vs. non-energy-aware systems, while cutting latency by 30–40%.
- Trained neural network models in PyTorch to infer adaptive, per-user QoS recommendations from network telemetry, replacing rule-based decision logic with data-driven inference.
- Designed a Gurobi-based optimization pipeline for energy-aware path selection and transmit-power control, jointly minimizing latency and energy under strict QoS constraints.
- Results published in *IEEE Networking Letters* (2025), co-authored with Futurewei researchers.

Ozyegin University, Istanbul, Turkey – Research Assistant Jan 2018 – Jul 2021

- Developed convex optimization algorithms for energy-efficient wireless systems, extending device battery lifetime by up to 110%.

Projects

AI Job Scout Agent (Python, Gemini API)

- Built a multi-agent job-search assistant: LLM reasoning over multiple job-data sources, embedding-based semantic ranking of postings, and a SQLite relational store with analytical queries for tracking and deduplication.
- Extended with agentic deployment patterns using Google’s Agent Development Kit (ADK) and Vertex AI (Google AI Agents Workshop, Jun 2026).

Education

PhD in Electrical Engineering, New Jersey Institute of Technology

Newark, NJ

Dissertation: Reliable Machine Learning and Information Retrieval under Uncertainty

Jan 2022 – May 2026

Selected Publications

Full list on [Google Scholar](#).

1. **S. Ghasvarianjahromi**, J. Barr, Y. Yakimenka, J. Kliever, “Context-Aware Search and Retrieval Under Token Erasure,” under review, 2026.
2. **S. Ghasvarianjahromi**, A. Kiani, A. Xiang, J. Kaippallimalil, T. Saboorian, N. Ansari, “AI/ML-based QoS Recommendations for Energy Optimization in 6G Networks,” *IEEE Netw. Lett.*, 2025.
3. M. Bakshi, **S. Ghasvarianjahromi**, Y. Yakimenka, A. Beemer, O. Kosut, J. Kliever, “VALID: A Validated Algorithm for Learning in Decentralized Networks with Possible Adversarial Presence,” *IEEE Intern. Symp. Info. Theo. (ISIT)*, 2024.

Professional Service

IEEE Reviewer – Wireless Communications Letters, Communications Letters, Transactions on Communications, Transactions on Vehicular Technology, Transactions on Wireless Communications, Elsevier Physical Communications